

Exercise 1

Results for n = 10:

```
f=@integrand;
a=0; % lower limit
b=1; % upper limit
n=10;
trapezoidal = segment_trapezoidal(f,a,b,n);
built_in_integral = integral(f,a,b);
simpson = simpson_one_third(f,a,b,n);
true_value = 0.602297;
percent_error_trapezoidal = 100*abs(true_value-trapezoidal)/true_value;
percent_error_integral = 100*abs(true_value-built_in_integral)/true_value;
percent_error_simpson = 100*abs(true_value-simpson)/true_value;
fprintf(['trapezoidal = %4.5f \n' ...
        'built_in_integral = %4.5f \n' ...
        'simpson = %4.5f \n' ...
        'percent_error_trapezoidal = %4.5f \n' ...
        'percent_error_integral = %4.5f \n' ...
        'percent_error_simpson = %4.5f \n'],trapezoidal, built_in_integral,simpson,percent_error_trapezoidal,
percent_error_integral,percent_error_simpson)
```

```
trapezoidal = 0.55980
built_in_integral = 0.60230
simpson = 0.57738
percent_error_trapezoidal = 7.05652
percent_error_integral = 0.00018
percent_error_simpson = 4.13712
```

Results for n = 100:

```
f=@integrand;
a=0; % lower limit
b=1; % upper limit
n=100;
trapezoidal = segment_trapezoidal(f,a,b,n);
built_in_integral = integral(f,a,b);
simpson = simpson_one_third(f,a,b,n);
true_value = 0.602297;
percent_error_trapezoidal = 100*abs(true_value-trapezoidal)/true_value;
percent_error_integral = 100*abs(true_value-built_in_integral)/true_value;
percent_error_simpson = 100*abs(true_value-simpson)/true_value;
fprintf(['trapezoidal = %4.5f \n' ...
        'built_in_integral = %4.5f \n' ...
        'simpson = %4.5f \n' ...
        'percent_error_trapezoidal = %4.5f \n' ...
        'percent_error_integral = %4.5f \n' ...
        'percent_error_simpson = %4.5f \n'],trapezoidal, built_in_integral,simpson,percent_error_trapezoidal,
percent_error_integral,percent_error_simpson)
```

```
'percent_error_simpson = %4.5f \n'],trapezoidal, built_in_integral,simpson,percent_error_tr
percent_error_simpson)
```

```
trapezoidal = 0.59911
built_in_integral = 0.60230
simpson = 0.60034
percent_error_trapezoidal = 0.52914
percent_error_integral = 0.00018
percent_error_simpson = 0.32446
```

Results for n = 1000:

```
f=@integrand;
a=0; % lower limit
b=1; % upper limit
n=1000;
trapezoidal = segment_trapezoidal(f,a,b,n);
built_in_integral = integral(f,a,b);
simpson = simpson_one_third(f,a,b,n);
true_value = 0.602297;
percent_error_trapezoidal = 100*abs(true_value-trapezoidal)/true_value;
percent_error_integral = 100*abs(true_value-built_in_integral)/true_value;
percent_error_simpson = 100*abs(true_value-simpson)/true_value;
fprintf(['trapezoidal = %4.5f \n' ...
        'built_in_integral = %4.5f \n' ...
        'simpson = %4.5f \n' ...
        'percent_error_trapezoidal = %4.5f \n' ...
        'percent_error_integral = %4.5f \n' ...
        'percent_error_simpson = %4.5f \n'],trapezoidal, built_in_integral,simpson,percent_error_tr
percent_error_simpson)
```

```
trapezoidal = 0.60205
built_in_integral = 0.60230
simpson = 0.60214
percent_error_trapezoidal = 0.04153
percent_error_integral = 0.00018
percent_error_simpson = 0.02560
```

```
function [I] = segment_trapezoidal(f,a,b,n)

I=0; % initial value of integral
h=(b-a)/n;
for i=0:n
    x= a + i*h; % goin from a to b
    if (i==0 || i==n)
        I = I + f(x);
    else % if is not 0 or n
        I = I + 2*f(x);
    end
end
I = I*h/2;
end

function [y] = integrand(x) % integrand function
```

```

y=x.^(0.1).*(1.2-x).*(1-exp(20.*(x-1)));
end

function [I] = simpson_one_third(f,a,b,n)
    if (mod(n,2)==0)
        I=0;
        h = (b-a)/n;
        for i =0:n
            x = a + i*h;
            if (i==0||i==n)
                I = I + f(x);
            elseif (mod(i,2)==1)
                I = I + 4*f(x);
            else
                I = I + 2*f(x);
            end
        end

        I = I*h/3;
    else
        disp("n has to be even")
    end
end
end

```